

Problem 1.0: Schrödinger Equation in Different Basis



As a warm-up question, starting from the time-dependent Schrödinger equation in operator form,

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle,$$

where $\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}})$, we will derive the “standard” Schrödinger equation in terms of the position-space and momentum-space wavefunctions. Conceptually, this is a trivial exercise, but it does require some elbow grease, such as much of what is to come in this course.

- a) Derive the time-dependent Schrödinger equation in position space (in 3 spatial dimensions), i.e., find the differential equation obeyed by the $\psi(\mathbf{x}, t) = \langle \mathbf{x} | \psi(t) \rangle$.
(Hint: use the completeness relation of the position basis, $\int d^3\mathbf{x} |\mathbf{x}\rangle \langle \mathbf{x}| = \hat{I}$, and the fact that the momentum operator in position space is $\hat{\mathbf{p}} = -i\hbar \nabla$. You can assume that the potential operator depends exclusively on powers of $\hat{\mathbf{x}}$, such that $V(\hat{\mathbf{x}}) |\mathbf{x}\rangle = V(\mathbf{x}) |\mathbf{x}\rangle$.)
- b) Repeat the same steps to derive the time-dependent Schrödinger equation in momentum space, i.e., find the *integro-differential* equation obeyed by $\phi(\mathbf{p}, t) = \langle \mathbf{p} | \psi(t) \rangle$.
(Hint: recall that the momentum and position kets are related by the Fourier transform, $\langle \mathbf{x} | \mathbf{p} \rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}$.)
- c) Now, using the Fourier transform relations between position and momentum space wavefunctions,

$$\psi(\mathbf{x}, t) = \frac{1}{(2\pi\hbar)^{3/2}} \int d^3\mathbf{p} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar} \phi(\mathbf{p}, t),$$

$$\phi(\mathbf{p}, t) = \frac{1}{(2\pi\hbar)^{3/2}} \int d^3\mathbf{x} e^{-i\mathbf{p}\cdot\mathbf{x}/\hbar} \psi(\mathbf{x}, t),$$

show that the two Schrödinger equations you derived in (a) and (b) are equivalent.

(Hint: start from the position space Schrödinger equation. Recall that the delta function can be written as $\delta(\mathbf{p} - \mathbf{p}') = \frac{1}{(2\pi\hbar)^3} \int d^3\mathbf{x} e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}/\hbar}$.)

Problem 1.1: Probability density and current



The time-dependent Schrödinger equation in 3D is

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{x}, t) = -\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{x}, t) + V(\mathbf{x}) \psi(\mathbf{x}, t).$$

Using the Schrödinger equation, we can calculate the probability density $\rho(\mathbf{x}, t) = |\psi(\mathbf{x}, t)|^2$, which gives the probability of finding the particle at an infinitesimally small region around $\mathbf{x}$ at time t . For a closed system, the probability density should satisfy the continuity equation:

$$\frac{\partial \rho(\mathbf{x}, t)}{\partial t} + \nabla \cdot \mathbf{j}(\mathbf{x}, t) = 0.$$

This is a statement of local conservation of probability, where $\mathbf{j}(\mathbf{x}, t)$ is the probability current density (how much probability is flowing through a unit area per unit time at position $\mathbf{x}$ and time t).

- a) Derive the standard expression for the probability current density:

$$\mathbf{j}(\mathbf{x}, t) = \frac{\hbar}{2mi} (\psi^*(\mathbf{x}, t) \nabla \psi(\mathbf{x}, t) - \psi(\mathbf{x}, t) \nabla \psi^*(\mathbf{x}, t)).$$

- b) Rewrite $\mathbf{j}(\mathbf{x}, t)$ in terms of the momentum operator $\hat{\mathbf{p}} = -i\hbar \nabla$.
- c) In the Schrödinger picture and using Dirac notation, find an expression for the current operator $\hat{j}(\mathbf{x})$ in a basis-independent way. Is this operator time-dependent? What about its expectation value? (hint: recall that $\psi(\mathbf{x}) = \langle \mathbf{x} | \psi \rangle$ and $\psi^*(\mathbf{x}) = \langle \psi | \mathbf{x} \rangle$).
- d) Now consider a polar form for the wave function: $\psi(\mathbf{x}, t) = \sqrt{\rho(\mathbf{x}, t)} e^{iS(\mathbf{x}, t)/\hbar}$, where $\rho(\mathbf{x}, t)$ is the probability density and $S(\mathbf{x}, t)$ is the phase of the wave function. Show that the probability current density can be written as

$$\mathbf{j}(\mathbf{x}, t) = \frac{\rho(\mathbf{x}, t)}{m} \nabla S(\mathbf{x}, t).$$

What does this mean for a *real* wavefunction $\psi(\mathbf{x})$?

- e) For a plane-wave solution of the form $\psi(\mathbf{x}, t) = \frac{1}{\sqrt{V}} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$, where V is some normalization volume (formally, $V \rightarrow \infty$ for a bona-fide plane wave), $\mathbf{k}$ is the wave vector, and ω is the angular frequency, compute the probability density $\rho(\mathbf{x}, t)$. Is it time-dependent? How about the probability current density $\mathbf{j}(\mathbf{x}, t)$?